NAME: Krunal Savaj

Project Proposal

Title

Incentive-Aware Multi-Objective Matching for Peer-to-Peer Commuter Carpooling

Name and Role

Krunal : Responsible for all Aspects

Communication Plan

Not Applicable

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1. Problem Statement and Motivation

Urban commuting systems experience congestion, high transportation costs, and underutilized private vehicle capacity. While ride-sharing platforms such as **Uber and Lyft** offer pooled ride options, their optimization frameworks are primarily designed for professional drivers and real-time demand rather than peer commuters with partially fixed routes.

In commuter-oriented peer-to-peer settings (e.g., university campuses or office corridors), participants are regular commuters with limited tolerance for detours. Effective ride matching must therefore simultaneously:

- Minimize additional detour time for drivers
- Increase economic benefit for participants
- Maximize overall match rate
- Reduce cancellation likelihood
- Maintain lower rider cost compared to solo alternatives

Traditional geographic threshold–based matching does not explicitly model trade-offs between detour cost, economic gain, and match probability. In this project, I propose to design and evaluate a machine learning–based multi-objective matching framework that predicts route compatibility and match success likelihood, and optimizes driver–rider pairing under competing objectives.

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2. Dataset Description

Primary Dataset

NYC Taxi Trip Record Data
Provided by the New York City Taxi and Limousine Commission (NYC TLC)

Official page:
https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page

Direct download directory (example – Yellow Taxi):
https://d37ci6vzurychx.cloudfront.net/trip-data/

The dataset includes:

- Pickup and drop-off timestamps
- Pickup and drop-off latitude/longitude
- Trip distance
- Trip duration
- Fare amount
- Passenger count

Optional Supplementary Dataset

Chicago Taxi Trips Dataset (City of Chicago Open Data):
https://data.cityofchicago.org/Transportation/Taxi-Trips/wrvz-psew

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Data Construction Plan

Because the datasets contain single-passenger trips **rather than shared rides**, I will construct a simulation framework:

1. Cluster trips into commuting time windows (e.g., 7–10 AM, 4–7 PM).
2. Define candidate driver–rider pairs based on spatial and temporal overlap.
3. Engineer features including:
   - Route overlap ratio
   - Detour distance
   - Detour time
   - Estimated incremental earning
   - Time window compatibility

This synthetic pairing framework enables controlled evaluation of matching strategies.

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4. Baseline Model

I will implement two baseline approaches:

1. Rule-Based Matching

- **Objective:** Provide a simple, interpretable benchmark to evaluate more advanced ML methods.

- **Method:**
  - Compute straight-line (Haversine) distance between driver route and rider pickup/drop points.

  - Accept matches only if detour distance is below a fixed threshold (e.g., 2–3 km).
  - Time window compatibility: rider pickup must fall within driver commute time ±15 minutes.
  - Cost sharing: fare split proportional to distance traveled.

- **Input Features:**
  - Driver start/end coordinates
  - Rider pickup/drop coordinates
  - Estimated detour distance/time

- **Output:**
  - Binary match feasibility (1 = feasible, 0 = infeasible)

- **Rationale:** Provides interpretable baseline and allows quick sanity checks for route compatibility.

2. Logistic Regression Baseline

- **Objective:** Introduce a simple predictive model that captures linear relationships between key features and match feasibility.

- **Method:**
  - Use the same candidate pairs as the rule-based method.
  - Train a logistic regression classifier to predict match feasibility.

- **Input Features:**
  - Detour distance
  - Detour time
  - Route overlap ratio
  - Time window overlap
  - Rider-driver distance from route midpoint

- **Output:** Probability of match success (acceptance probability)

- **Rationale:** Provides a quantitative benchmark for comparison with non-linear, multi-objective models.

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5. Advanced Model

The advanced model focuses on **multi-objective prediction** and ranking of driver–rider pairs.

Core Model

- **Algorithm:** Gradient Boosted Trees (XGBoost)

- **Objectives:**
  1. Predict detour cost (continuous regression)

2. Predict match probability (classification)
3. Compute a composite score for multi-objective matching

- **Input Features:**
  - Driver route coordinates (start, end, waypoints)
  - Rider pickup/drop coordinates
  - Detour distance and estimated time deviation
  - Route similarity metrics (e.g., fraction of route overlapped)
  - Time window overlap
  - Rider trip length

- **Output:**
  - Predicted detour time (minutes)
  - Probability of driver accepting match
  - Multi-objective match score

Rationale

- Captures non-linear relationships between detour, time, and likelihood of acceptance.
- Enables multi-objective optimization for realistic commuter scenarios.
- Provides a flexible scoring framework that can be extended.

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6. Evaluation Metrics

Model-Level

Regression: - Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)

Classification: - Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

System-Level Simulation

- Match rate (%)
- Average driver detour
- Average driver earnings increase
- Average rider cost reduction
- Trade-off curves (earnings vs detour)

Baseline vs advanced model comparison will be performed.

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7. Planned Timeline (Now – End of April)

Week 1

- Collect and clean NYC/Chicago taxi datasets
- Perform exploratory data analysis (EDA)
- Cluster trips into commuting time windows
- Engineer route similarity features, detour distance, detour time, and time window compatibility

Week 2

- Implement rule-based matching system
- Implement logistic regression baseline
- Evaluate baseline models on initial synthetic driver–rider pairs
- Analyze feature importance and baseline performance

Week 3

- Implement Gradient Boosted Trees (XGBoost) for multi-objective matching
- Train regression for detour prediction and classification for match probability
- Construct multi-objective scoring function combining detour, earnings, and acceptance probability
- Perform initial hyperparameter tuning and validation

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8. Areas of Uncertainty / Assistance Requested

- Since my match labels are simulated, what strategies would you suggest for validating that these labels are realistic and suitable for ML training? Are there best practices for train/test splitting or sensitivity analysis for synthetic labels?
- How can I ensure that my evaluation methodology is rigorous and scientifically credible, given that the dataset and labels are simulated? For example, how should I document assumptions, benchmark against baselines, and present robustness analysis?